

Module 5: Matrix Algebra

Ph.D. Math Camp

1 Linear Models and Matrix Algebra

Thus far when solving for unknowns (e.g., P^*, Q^*) we have stuck with two equations and two unknowns.

Including more unknowns and equations makes solutions cumbersome. In order to handle large systems of simultaneous **linear** equations we need *matrix algebra*.¹

Recall we had a system of two linear equations when we analyzed supply and demand

$$c_1q + c_2p = b$$

$$\gamma_1q - \gamma_2p = d$$

In general, a system of n linear equations in n variables $(x_1, x_2, x_3, \dots, x_n)$ can be arranged in the following format

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = d_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = d_2$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = d_m$$

Here, the variable x_1 appears only within the leftmost column, and in general, x_j appears only in the j th column.

¹Can also apply transformations $y = ax^b \implies \ln y = \ln a + b \ln x$

a_{ij} represents the coefficient appearing in the i th equation that is attached to the j th variable. For example, a_{21} is the coefficient in the second equation, attached to x_1 . Here, d_i is unattached to any variable and represents the constant term in the i th equation.

1.1 Matrices as Arrays

There are three components in the system of equations (above).

- The set of coefficients a_{ij} .
- The set of variables $x_1, x_2, x_3, \dots, x_n$
- The constant terms $d_1, d_2, \dots, d_m$

If we arrange the three sets as rectangular arrays and label them as A , x , and d then we have

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \quad d = \begin{bmatrix} d_1 \\ d_2 \\ \dots \\ d_m \end{bmatrix}$$

As an example, given the linear system of equations

$$6x_1 + 3x_2 + x_3 = 22$$

$$x_1 + 4x_2 - 2x_3 = 12$$

$$4x_1 - x_2 + 5x_3 = 10$$

we can write

$$A = \begin{bmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad d = \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix}$$

Each of these three arrays constitutes a matrix.

Matrix: A rectangular array of numbers, parameters, or variables. The numbers of the array, referred to as *elements* of the matrix, are usually enclosed in brackets.

Note that the elements are separated by spaces, not commas. As a shorthand, the array in matrix A can be written more simply as

$$A = [a_{ij}], \quad \begin{pmatrix} i = 1, 2, \dots, m \\ j = 1, 2, \dots, n \end{pmatrix}$$

1.2 Vectors as Special Matrices

Let's start with some terminology

- Scalar: A real number
- Vector: Ordered collection of n elements (called components or coordinates)

If a vector has n elements it is said to be of dimension n i.e. $\begin{bmatrix} 3 \\ 7 \\ 6 \end{bmatrix}$. This is a column vector of dimension 3. You can have a row vector of dimension 4 e.g., $[1 \ 7 \ 2 \ 6]$.

The number of rows and the number of columns in a matrix together define the *dimension* of the matrix. Since matrix A contains m rows and n columns, it is said to be of dimension $m \times n$ (read “m by n”). To be consistent, the row number should always precede the column number.

In the special case where $m = n$, the matrix is called a *square matrix*. Thus, matrix A is a 3×3 square matrix.

Some matrices may contain only one column, such as x and d . These matrices are called *column* vectors.

The dimension of x is $n \times 1$ and d is $m \times 1$.

If we arranged the variables x_j in a horizontal array, the result should be a $1 \times n$ matrix, which is called a *row vector*. For notation purposes, a row vector is often distinguished from a column vector by the use of a primed symbol:

$$x' = [x_1 \quad x_2 \quad x_3 \quad \dots \quad x_n]$$

With matrices defined, we can express an equation system as follows

$$Ax = d$$

No matter how extensive the system is, we can express it this way.

1.3 Matrix Operations

Note that **equality** means that two matrices are equal iff they have the same dimensions and have identical elements.

$$\begin{bmatrix} 4 & 3 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 2 & 0 \end{bmatrix} \neq \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$$

As another example, if $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$ means that $x = 7$ and $y = 4$.

Addition and Subtraction

Two matrices can be added iff they have the same dimension. When this dimensional requirement is met, the matrices are said to be *conformable for addition*.

In that case, the addition of $A = [a_{ij}]$ and $B = [b_{ij}]$ is defined as the addition of each pair of corresponding elements.

Generally, we can state the rule as $[a_{ij}] + [b_{ij}] = [c_{ij}]$ where $c_{ij} = a_{ij} + b_{ij}$. Note that $[c_{ij}]$ will have the same dimension as the matrices $[a_{ij}]$ and $[b_{ij}]$.

The **subtraction** operation $A - B$ can similarly be defined if and only if A and B have the same dimension. The operation entails the results

$$[a_{ij}] - [b_{ij}] = [d_{ij}] \quad \text{where} \quad d_{ij} = a_{ij} - b_{ij}$$

Scalar Multiplication

To multiply a matrix by a number– or in matrix-algebra terminology, by a *scalar*– is to multiply **every** element of that matrix by the given scalar. Thus, one “scales up (or down)”

Examples:

(i) Add A+B. $A = \begin{bmatrix} 4 & 9 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$ Add the corresponding elements

$$A + B = \begin{bmatrix} 4+2 & 9+0 \\ 2+0 & 1+7 \end{bmatrix} = \begin{bmatrix} 6 & 9 \\ 2 & 8 \end{bmatrix}$$

(ii) A-B. $\begin{bmatrix} 19 & 3 \\ 2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 8 \\ 1 & 3 \end{bmatrix}$

$$A - B = \begin{bmatrix} 19-6 & 3-8 \\ 2-1 & 0-3 \end{bmatrix} = \begin{bmatrix} 13 & -5 \\ 1 & -3 \end{bmatrix}$$

(iii) $7 \begin{bmatrix} 3 & -1 \\ 0 & 5 \end{bmatrix}$

(iv) $\frac{1}{2} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

(v) $A + (-1)B. \quad A = \begin{bmatrix} 4 & 9 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$

2 Matrix Operations (Continued)

2.1 Multiplication of Matrices

The multiplication of two matrices is contingent upon the satisfaction of a different dimensional requirement.

Suppose that, given two matrices A and B , we want to find the product AB . The *conformability condition* for multiplication is that the *column* dimension of A (the “lead” matrix in AB) must equal the *row* dimension of B (the “lag” matrix). For instance,

$$\underset{(1 \times 2)}{A} = [a_{11} \ a_{12}] \quad \underset{(2 \times 3)}{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

the product AB then is defined, since A has *two columns* and B has *two rows*. You can check this quickly by checking the second number in the dimension indicator for A and the first number in the dimension indicator for B .

Conversely, BA is not defined.

Generally, if A is of dimension $m \times n$ and B is of dimension $p \times q$, the matrix product will be defined iff $n = p$.

If defined, the product of AB will have dimension $m \times q$. In this example, AB will have dimension 1×3 .

Thus, the resulting matrix (call it C) will take the form

$$AB = C = [c_{11} \ c_{12} \ c_{13}]$$

c_{ij} is defined as a sum of the products, to be computed from the elements in the i th row of A , and those in the j th column of B .

For instance, to find c_{11} we take the first row in A and the first column in B and pair the elements sequentially, multiply out each pair, and take the sum of the resulting products, to get

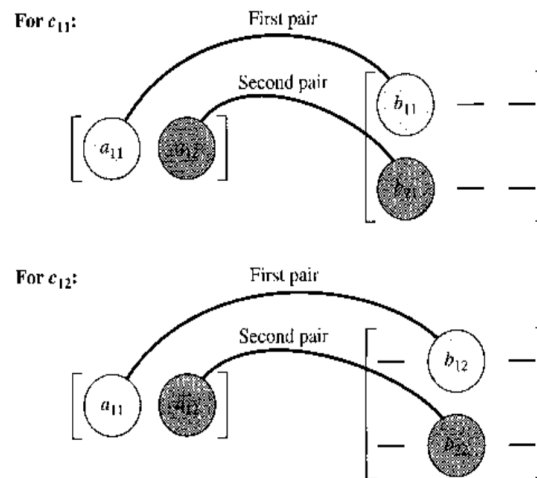
$$c_{11} = a_{11}b_{11} + a_{12}b_{21}$$

Similarly, for c_{12}

$$c_{12} = a_{11}b_{12} + a_{12}b_{22}$$

And for $c_{13} = a_{11}b_{13} + a_{12}b_{23}$

Figure 1: Matrix Multiplication



The multiplication procedure illustrated in the figure can also be described using the concept of *inner product* of two vectors. Given two vectors u and v with n elements each, say, $(u_1, u_2, u_3, \dots, u_n)$ and $(v_1, v_2, \dots, v_n)$, arranged as *either* two rows or as two columns or as one row and one column, their inner product written as $u \cdot v$ is defined as

$$u \cdot v = u_1v_1 + u_2v_2 + \dots + u_nv_n$$

Note that the number of elements is the same.

Examples:

(i) Find AB . $A = \begin{bmatrix} 1 & 3 \\ 2 & 8 \\ 4 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$

$$AB = C = \begin{bmatrix} c_{11} \\ c_{21} \\ c_{31} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} \\ a_{21}b_{11} + a_{22}b_{21} \\ a_{31}b_{11} + a_{32}b_{21} \end{bmatrix} = \begin{bmatrix} 1(5) + 3(9) \\ 2(5) + 8(9) \\ 4(5) + 0(9) \end{bmatrix} = \begin{bmatrix} 32 \\ 82 \\ 20 \end{bmatrix}$$

(ii) Find AB . $A = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 3 \\ 4 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -\frac{1}{5} & \frac{3}{10} \\ -1 & \frac{1}{5} & \frac{7}{10} \\ 0 & \frac{2}{5} & -\frac{1}{10} \end{bmatrix}$

$$AB = \begin{bmatrix} 3(0) + (-1)(-1) + 2(0) & 3(-\frac{1}{5}) + (-1)\frac{1}{5} + 2(\frac{2}{5}) & 3(\frac{3}{10}) + (-1)\frac{7}{10} + 2(-\frac{1}{10}) \\ 1(0) + 0(-1) + 3(0) & 1(-\frac{1}{5}) + 0(\frac{1}{5}) + 3(\frac{2}{5}) & 1(\frac{3}{10}) + 0(\frac{7}{10}) + 3(-\frac{1}{10}) \\ 4(0) + 0(-1) + 2(0) & 4(-\frac{1}{5}) + 0(\frac{1}{5}) + 2(\frac{2}{5}) & 4(\frac{3}{10}) + 0(\frac{7}{10}) + 2(-\frac{1}{10}) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This square matrix with 1s in its *principal diagonal* and 0s everywhere else exemplifies an important type of matrix known as the *identity matrix*.

(iii) Let us revisit the following system of equations. Write out as a product of matrices.

$$6x_1 + 3x_2 + x_3 = 22$$

$$x_1 + 4x_2 - 2x_3 = 12$$

$$4x_1 - x_2 + 5x_3 = 10$$

$$Ax = \begin{bmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{bmatrix}_{(3 \times 3)} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{(3 \times 1)} = \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix}_{(3 \times 1)} = d$$

Note that to use the matrix notation $Ax = d$, it is necessary, because of the conformability condition, to arrange the x_j s into a column vector, even though they are listed in horizontal order.

We can also use the summation shorthand to write out the matrix multiplication discussed above.

$$c_{11} = a_{11}b_{11} + a_{12}b_{21} = \sum_{k=1}^2 a_{1k}b_{k1} \quad (1)$$

$$c_{12} = a_{11}b_{12} + a_{12}b_{22} = \sum_{k=1}^2 a_{1k}b_{k2} \quad (2)$$

$$c_{13} = a_{11}b_{13} + a_{12}b_{23} = \sum_{k=1}^2 a_{1k}b_{k3} \quad (3)$$

In each case, the first subscript of c_{1j} is reflected in the first subscript of a_{1k} , and the second subscript of c_{1j} is reflected in the second subscript of b_{kj} in the Σ expression. The index k is a dummy subscript and serves to indicate which particular pair of elements is being multiplied.

Extending this to the multiplication of an $m \times n$ matrix $A = [a_{ik}]$ and an $n \times p$ matrix $B = [b_{kj}]$, we may now write the elements of the $m \times p$ product matrix $AB = C = [c_{ij}]$ as

$$c_{11} = \sum_{k=1}^n a_{1k}b_{k1} \quad c_{12} = \sum_{k=1}^n a_{1k}b_{k2} \dots$$

or more generally,

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} \quad \left(\begin{bmatrix} i = 1, 2, \dots, m \\ j = 1, 2, \dots, p \end{bmatrix} \right)$$

2.2 Additional Vector Operations

An $m \times 1$ column vector u , and a $1 \times n$ row vector v' , yield a product matrix uv' of dimension $m \times n$. Suppose $u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $v' = [1 \ 4 \ 5]$

$$\Rightarrow uv' = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \times [1 \ 4 \ 5] = \begin{bmatrix} 3 \times 1 & 3 \times 4 & 3 \times 5 \\ 2 \times 1 & 2 \times 4 & 2 \times 5 \end{bmatrix} = \begin{bmatrix} 3 & 12 & 15 \\ 2 & 8 & 10 \end{bmatrix}$$

On the other hand if we were interested in multiplying $u' = [3 \ 4]$ and $v = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$

$$u'v = [3(9) + 4(7)] = [55]$$

As another example suppose $u' = [3 \ 6 \ 9]$. Find $u'u$.

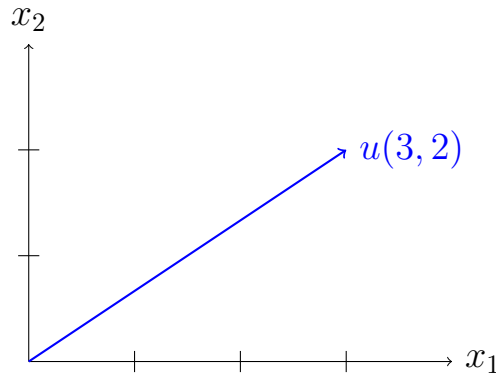
$$u'u = [3 \ 6 \ 9] \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = [(3)^2 + (6)^2 + (9)^2] = [126]$$

Note that $u'u$ is not the same as uu' . The latter will result in a matrix larger than 1×1 .

2.3 Geometric Interpretation of Vector Operations

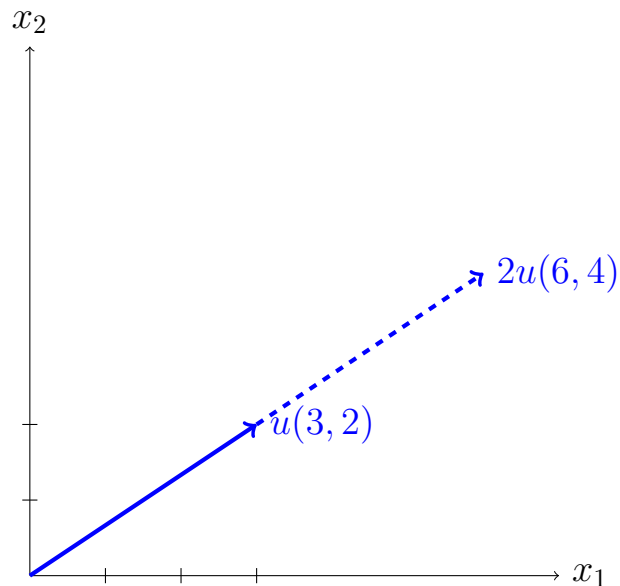
A column or row vector can also be viewed as a point in space. Suppose we plot the point (3,2) in a 2-space and label it u . This is the geometric counterpart of the vector $u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ or the vector $u' = [3 \ 2]$. Both indicate the same ordered pair.

If an arrow is drawn from the point of the origin (0,0) to the point u , it will specify the unique straight route by which to reach the destination point u from the point of origin. Such an arrow, with a definite length and direction, is called a *radius vector*.



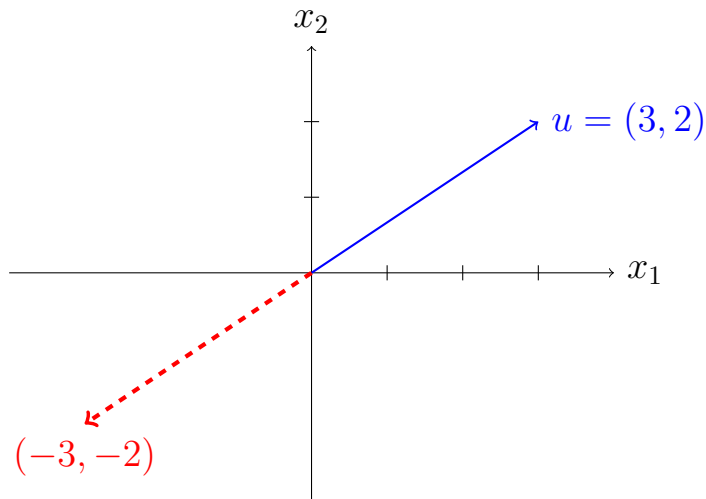
We can now get more insights from scalar multiplication, addition and subtraction of vectors, and linear combinations of vectors.

For instance, if we plot $2u = 2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$

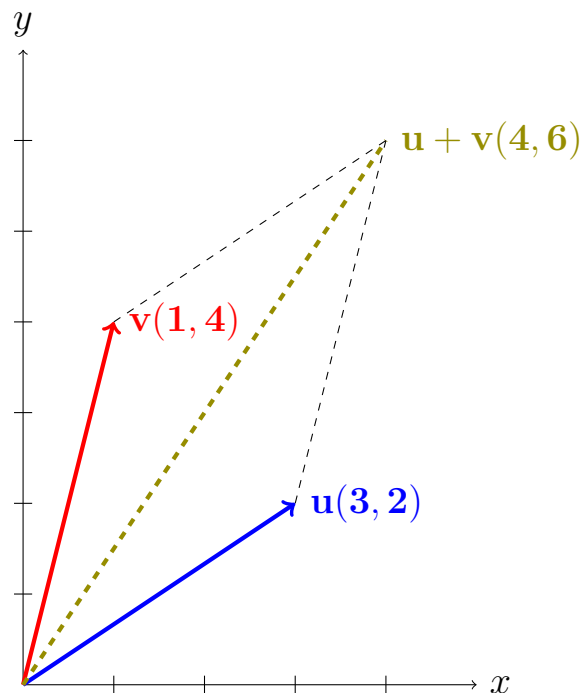


Multiplication of a vector u by any scalar k will produce an overlapping arrow the arrowhead will be relocated, unless $k = 1$. If $k = 0$, the arrow will shrink into the point of the origin representing a null vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

A negative scalar multiplier will reverse the direction of the arrow.



We can also add vectors. Consider $v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ and $u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ the sum $u + v = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$



We can do the same with a vector difference.

2.4 Linear Dependence

A set of vectors $v_1, \dots, v_n$ is said to be *linearly dependent* if (and only if) any one of them can be expressed as a linear combination of the remaining vectors; otherwise,

they are *linearly independent*.

1. For example, take the vectors $v_1 = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$. These are linearly dependent because v_3 is a linear combination of v_1 and v_2 :

$$3v_1 - 2v_2 = \begin{bmatrix} 6 \\ 21 \end{bmatrix} - \begin{bmatrix} 2 \\ 16 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} = v_3$$

We can also express this equation as $3v_1 - 2v_2 - v_3 = 0$. Where $0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is a zero vector.

2. Another example of linearly dependent vectors: $v'_1 = [5 \ 12]$ and $v'_2 = [10 \ 24]$.

$$2v'_1 = 2[5 \ 12] = [10 \ 24] = v'_2$$

Note again that this equation can be written as $2v'_1 - v'_2 = 0'$. $0'$ is a null row vector

We can thus redefine linear dependence as follows: A set of m -vectors $v_1, \dots, v_n$ is linearly dependent if and only if there exists a set of scalars $k_1, \dots, k_n$ (not all zero) such that

$$\sum_{i=1}^n k_i v_i = \underset{m \times 1}{0}$$

If $k_i = 0$ is the only set of values that can satisfy this equality then we say that these vectors are *linearly independent*.

We can also see this geometrically. u and $2u$ are dependent, while u and v are independent since we cannot express one as a multiple of the other. Their arrows do not lie on a single straight line.

2.5 Vector Space

It turns out that by extending, shortening, and reversing a given independent vectors u and v and combining them, we can generate an infinite number of new vectors. We call this the two-dimensional *vector space*.

Two linearly independent vectors u and v can be a basis for 2-space. Consider two vectors $[1 \ 0]$ and $[0 \ 1]$, which are called unit vectors. The first one plots an arrow lying along the horizontal axis, and the second an arrow lying along the vertical axis. Since they are linearly independent they can serve as the basis for 2-space.

By analogy, the three-dimensional vector space is the totality of 3-vectors, and it must be spanned by exactly three linearly independent 3-vectors. Consider the set of three unit vectors:

$$e_1 \equiv \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad e_2 \equiv \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad e_3 \equiv \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

These arrows lie on the three axes of the 3-space. We can generate the entire 3-space from these vectors. For instance, take the vector $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. This vector can be considered the linear combination $e_1 + 2e_2 + 2e_3$.

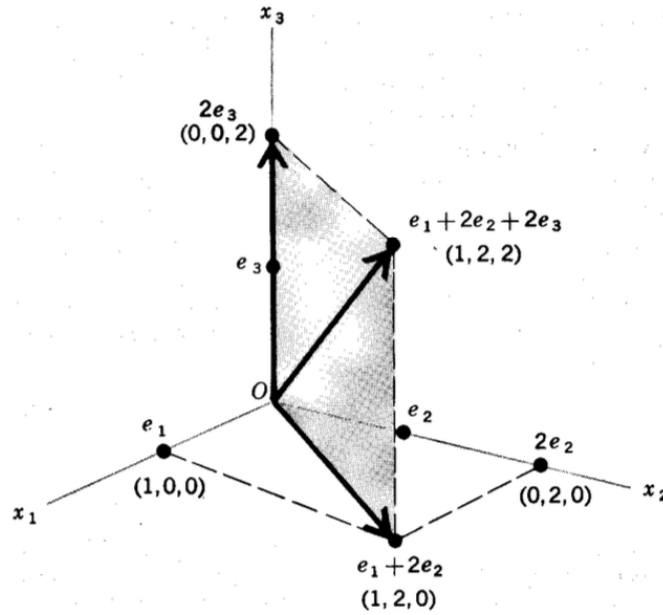
We can extend this to n -space (although not graphable). This n -space is sometimes referred to as *Euclidean n -space*.

Distance Between Vectors

For any pair of vectors u and v in a given space, the distance from u to v is some real-valued function

$$d = d(u, v)$$

Figure 2: Vector in 3-space



With the following properties:

1. when u and v coincide, the distance is zero
2. when the two points are distinct, the distance from u to v and the distance from v to u are represented by an identical positive real number
3. the distance between u and v is never longer than the distance from u to w (a point distinct from u and v) plus the distance from w to v .

We can summarize as follows:

$$d(u, v) = 0 \quad (\text{for } u = v)$$

$$d(u, v) = d(v, u) > 0 \quad (\text{for } u \neq v)$$

$$d(u, v) \leq d(u, w) + d(w, v) \quad (\text{for } w \neq u, v)$$

The last property is known as the *triangle inequality* because the three points u, v, w will usually define a triangle.

Distance in Euclidean space is defined as follows

$$d(u, v) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$$

For example, the distance between points (6,4) and (3,2) can be calculated as follows

$$\sqrt{(6 - 3)^2 + (4 - 2)^2} = \sqrt{3^2 + 2^2} = \sqrt{13}$$

2.6 Laws of Operations

Addition

- Commutative Law: $A + B = B + A$
- Associative Law: $(A + B) + C = A + (B + C)$

Given $v_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $v_2 = \begin{bmatrix} 9 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, we find that

$$(v_1 + v_2) - v_3 = \begin{bmatrix} 12 \\ 5 \end{bmatrix} - \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

Which is equal to

$$v_1 + (v_2 - v_3) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 7 \\ -4 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

Multiplication

- Matrix multiplication is **not** commutative

$$AB \neq BA$$

If AB is defined, that does not mean BA is defined. Even if both are, the general rule applies $AB \neq BA$.

Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 6 & 7 \end{bmatrix}$

$$AB = \begin{bmatrix} 1(0) + 2(6) & 1(-1) + 2(7) \\ 3(0) + 4(6) & 3(-1) + 4(7) \end{bmatrix} = \begin{bmatrix} 12 & 13 \\ 24 & 25 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0(1) + (-1)(3) & 0(2) - 1(4) \\ 6(1) + 7(3) & 6(2) + 7(4) \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 27 & 40 \end{bmatrix}$$

There are some interesting exceptions I will discuss shortly. Also, note that scalar multiplication of a matrix does obey the commutative law

$$kA = Ak$$

- Associative Law: $(AB)C = A(BC) = ABC$.

In forming the product ABC , the conformability condition must naturally be satisfied by each *adjacent* pair of matrices. If A is $m \times n$ and C is $p \times q$, then conformability requires B to be $n \times p$.

$$\begin{matrix} A & B & C \\ (m \times n) & (n \times p) & (p \times q) \end{matrix}$$

Examples: Find $C = AB$

$$(i) \ A = \begin{bmatrix} 12 & 14 \\ 20 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 9 \\ 0 & 52 \end{bmatrix}$$

$$(ii) \ A = \begin{bmatrix} 4 & 7 \\ 9 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 8 & 5 \\ 2 & 6 & 7 \end{bmatrix}$$

$$(iii) \ A = \begin{bmatrix} 7 & 11 \\ 2 & 9 \\ 10 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 12 & 4 & 5 \\ 3 & 6 & 1 \end{bmatrix}$$

$$(iv) \ A = \begin{bmatrix} 4 & 7 \\ 9 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(v) \quad A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 7 \\ 9 & 1 \end{bmatrix}$$

2.7 Identity Matrices

An identity matrix is a **square** matrix with 1s in its principal diagonals and 0s everywhere else. It is denoted by the symbol I or I_n . Where n indicates the dimension.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The importance of this special type of matrix lies in the fact that it plays a role similar to that of the number 1 in scalar algebra. For any number a , we have $1(a) = a(1) = a$. Similarly, for any matrix A , we have

$$IA = AI = A$$

As an example, suppose $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix}$

$$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix} = A$$

Similarly,

$$AI = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix} = A$$

Because A is 2×3 , what we would call *premultiplication* and *postmultiplication* of A by I would call for identity matrices of different dimensions to abide by conformability.

The special nature of the identity matrices makes it possible, during the multiplication process, to *insert* or *delete* an identity matrix without affecting the matrix product. For instance,

$$AIB = \begin{matrix} A & I & B \\ (m \times n) & (n \times n) & (n \times p) \end{matrix} = (AI)B = \begin{matrix} A & B \\ (m \times n) & (n \times p) \end{matrix}$$

which shows that the presence or absence of I does not affect the product.

An interesting case also occurs when $A = I_n$

$$AI_n = (I_n)^2 = I_n$$

The identity matrix squared is equal to itself. Generally, an identity matrix remains unchanged when it is multiplied by itself any number of times.

2.8 Null Matrices

Just as an identity matrix I plays the role of the number 1, a *null matrix*– or *zero matrix*– denoted by 0 , plays the role of the number 0. A null matrix is simply a matrix whose elements are all zero. This matrix is not restricted to being a square. Thus we can write

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

As an example,

$$A + 0 = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = A$$

However,

$$\begin{matrix} A & 0 \\ (2 \times 3) & (3 \times 1) \end{matrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Note that the resulting null matrix is 2×1 and not 3×1

Null matrices observe the following rules regarding multiplication and addition

$$\begin{matrix} A \\ (m \times n) \end{matrix} + \begin{matrix} 0 \\ (m \times n) \end{matrix} = \begin{matrix} 0 \\ (m \times n) \end{matrix} + \begin{matrix} A \\ (m \times n) \end{matrix} = \begin{matrix} A \\ (m \times n) \end{matrix}$$

$$\begin{matrix} A \\ (m \times n) \end{matrix} \begin{matrix} 0 \\ (n \times p) \end{matrix} = \begin{matrix} 0 \\ (m \times p) \end{matrix} \quad \text{and} \quad \begin{matrix} 0 \\ (q \times m) \end{matrix} \begin{matrix} A \\ (m \times n) \end{matrix} = \begin{matrix} 0 \\ (q \times n) \end{matrix}$$

A Couple More Differences from Scalar Algebra

- We have seen that $AB \neq BA$
- In scalar algebra $ab = 0 \implies a = 0 \text{ or } b = 0$. This is not so in matrix multiplication.

$$AB = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 4 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

- For scalars, the equation $cd = ce \ (c \neq 0) \implies d = e$. This does not always hold for matrices. Given

$$C = \begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \quad E = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}$$

We find that

$$CD = CE = \begin{bmatrix} 5 & 8 \\ 15 & 24 \end{bmatrix}$$

even though $D \neq E$.

2.9 Transposes and Inverses

When the rows and columns of a matrix A are interchanged— so that its first row becomes the first column, and vice versa— we obtain the *transpose* of A . We denote this as A' or A^T .

For example, given $\begin{matrix} A \\ (2 \times 3) \end{matrix} = \begin{bmatrix} 3 & 8 & -9 \\ 1 & 0 & 4 \end{bmatrix}$ and $\begin{matrix} B \\ (2 \times 2) \end{matrix} = \begin{bmatrix} 3 & 4 \\ 1 & 7 \end{bmatrix}$

We can interchange the rows and columns to get $A'_{(3 \times 2)} = \begin{bmatrix} 3 & 1 \\ 8 & 0 \\ -9 & 4 \end{bmatrix}$ and $B^T_{(2 \times 2)} = \begin{bmatrix} 3 & 1 \\ 4 & 7 \end{bmatrix}$

By definition, a matrix A is $m \times n$; its transpose A' must be $n \times m$. However, an $n \times n$ square matrix possesses a transpose with the same dimension.

Example: Find the transpose of each

$$(i) C = \begin{bmatrix} 9 & -1 \\ 2 & 0 \end{bmatrix}$$

$$(ii) D = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 3 & 7 \\ 4 & 7 & 2 \end{bmatrix}$$

In each of these examples, the dimension of each transpose is identical to that of the original matrix. In D' , we also note the remarkable result that D' inherits not only the dimension of D but also the original array of elements.

The fact that $D' = D$ is the result of the symmetry of the elements with reference to the principal diagonal. The principal diagonal in D is like a mirror.

The matrix D exemplifies the special class of square matrices known as *symmetric matrices*. Another example of this is the identity matrix, $I' = I$.

2.10 Properties of Transposes

- $(A')' = A$
- $(A + B)' = A' + B'$. Transpose of the sum is the sum of the transpose.

- $(AB)' = B'A'$. The transpose of the product is the product of the transpose in reverse order.

Examples:

1. Show that $(A + B)' = A' + B'$ for $A = \begin{bmatrix} 4 & 1 \\ 9 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 7 & 1 \end{bmatrix}$

2. Show that $(AB)' = B'A'$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 6 & 7 \end{bmatrix}$

2.11 Inverses and Their Properties

We can always derive the transpose of a matrix. However, its *inverse* matrix may or may not exist. The inverse of a matrix A , denoted by A^{-1} , is defined only if A is a square matrix, in which case the inverse is the matrix that satisfies the condition:

$$AA^{-1} = A^{-1}A = I$$

Whether you pre-or-post multiply by A^{-1} , the product will be the identity matrix.

Properties:

- Not every square matrix has an inverse— squareness is *necessary* but not *sufficient*. If a square matrix has an inverse, it is said to be *nonsingular*. If A has no inverse, it is called *singular*.

- If A^{-1} exists, then the matrix A can be regarded as the inverse of A^{-1} . They are inverses of each other.
- If an inverse exists, it is unique. We can show this. Suppose B is the inverse of A

$$AB = BA = I$$

Now suppose that there is another matrix C such that $AC = CA = I$.

If we premultiply $AB = I$ by C we find that

$$CAB = CI \implies CAB = C$$

Since we assumed $C = A^{-1} \implies CA = I$

$$IB = C \quad \text{or} \quad B = C$$

Thus, C is actually B . Therefore, we can speak of *an* (single) inverse of A .

- $(A^{-1})^{-1} = A$. The inverse of an inverse is the original matrix.
- $(AB)^{-1} = B^{-1}A^{-1}$. The inverse of the product is the product of the inverses.

Suppose that C is the inverse of $AB \implies C(AB) = I$. If we postmultiply both sides by $B^{-1}A^{-1}$

$$CABB^{-1}A^{-1} = IB^{-1}A^{-1}(= B^{-1}A^{-1})$$

$$\implies CA(BB^{-1})A^{-1} = B^{-1}A^{-1} \implies CAA^{-1} = B^{-1}A^{-1} \implies C = B^{-1}A^{-1}$$

- $(A')^{-1} = (A^{-1})'$. The inverse of the transpose is the transpose of the inverse.

Given A' , let us find the inverse, $D = (A')^{-1}$. By definition, we then have $DA' = I$.

$$(AA^{-1})' = I' = I$$

Thus, $DA' = (AA^{-1})' \implies DA' = (A^{-1})'A'$. Note that $(AA^{-1})' = I' = I$.

Postmultiplying both sides by $(A')^{-1}$, we obtain

$$DA'(A')^{-1} = (A^{-1})'A'(A')^{-1}$$

$$DI = (A^{-1})'$$

2.12 Solution of Linear-Equation System

We can now apply the concept of an inverse matrix to a simultaneous system of equations. Recall our example,

$$6x_1 + 3x_2 + x_3 = 22$$

$$x_1 + 4x_2 - 2x_3 = 12$$

$$4x_1 - x_2 + 5x_3 = 10$$

we can write

$$A = \begin{bmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad d = \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix}$$

If the inverse of the matrix A exists, the premultiplication of both sides yields

$$A^{-1}Ax = A^{-1}d \implies \underset{(3 \times 1)}{x} = \underset{(3 \times 3)}{A^{-1}} \underset{(3 \times 1)}{d}$$

The LHS is a column vector of variables and the RHS a product of known numbers. This allows us to find the solution for the x -values.

Since A^{-1} is unique, $A^{-1}d$ must be a unique vector of solutions. For this example

$$A^{-1} = \frac{1}{52} \begin{bmatrix} 18 & -16 & -10 \\ -13 & 26 & 13 \\ -17 & 18 & 21 \end{bmatrix}$$

Thus,

$$\begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{bmatrix} = \frac{1}{52} \begin{bmatrix} 18 & -16 & -10 \\ -13 & 26 & 13 \\ -17 & 18 & 21 \end{bmatrix} \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

Plugging these values into the system should satisfy each equation.

3 Conditions for Non-Singularity

Recall that a given coefficient matrix A can have an inverse (i.e., be non-singular) only if it is square. This condition is *necessary but not sufficient*.

A sufficient condition for the non-singularity of a matrix is that its **rows** be **linearly independent** (or its columns, which is the same thing).

Together,

1. Squareness
2. Linear independence

imply non-singularity.

3.1 Rows of a Matrix as Vectors

Recall that an $n \times n$ coefficient matrix A can be considered as an ordered set of row vectors,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} v'_1 \\ v'_2 \\ \vdots \\ v'_n \end{bmatrix}$$

or a column vector whose elements are themselves row vectors. Here,

$$v'_i = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}, \quad i = 1, 2, \dots, n.$$

For the rows to be linearly independent, *none must be a linear combination of the rest*.

Row linear independence requires that the only set of scalars k_i which can satisfy

$$\sum_{i=1}^n k_i v'_i = 0$$

be $k_i = 0$ for all i .

Example: Consider

$$A = \begin{bmatrix} 3 & 4 & 5 \\ 0 & 1 & 2 \\ 6 & 8 & 10 \end{bmatrix} = \begin{bmatrix} v'_1 \\ v'_2 \\ v'_3 \end{bmatrix}$$

Since

$$\begin{bmatrix} 6 & 8 & 10 \end{bmatrix} = 2 \begin{bmatrix} 3 & 4 & 5 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$$

the third row can be expressed as a linear combination of the first two, i.e., $v'_3 = 2v'_1 + 0v'_2$. Thus, the rows are *not* linearly independent.

We may also express this as

$$2v'_1 + 0v'_2 - v'_3 = \begin{bmatrix} 6 & 8 & 10 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 6 & 8 & 10 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

As long as the set of scalars k_i that led to the zero vector are not all zero, the rows are **linearly dependent**.

3.2 Why Do We Need Linear Independence?

You may recall that usually you hear that you need “as many equations as unknowns” to solve a system (e.g., two equations and two unknowns). However, the equations must be *functionally independent*. This is why we work with a square ($n \times n$) matrix: n *independent* equations and n unknowns together give a **unique solution**.

Example: Take

$$\begin{bmatrix} 10 & 4 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

Note that $v'_1 = 2v'_2$, which implies dependence. We don't know d_1 or d_2 , but there are two possibilities:

1. $d_1 = 2d_2$
2. $d_1 \neq 2d_2$

Consider (1), say $d_1 = 12$, $d_2 = 6$. The equations are consistent but linearly dependent. The first equation is the second equation times 2, so one equation is redundant. This is actually a single equation,

$$5x_1 + 2x_2 = 6,$$

with an infinite number of solutions that will satisfy both equations. There is no unique x_1^*, x_2^* : both equations describe the *same line*.

Consider (2), say $d_1 = 12$, $d_2 = 0$. The equations are *inconsistent*. If $10x_1 + 4x_2 = 12$ is true, then $5x_1 + 2x_2 = 6$ must be true — but $d_2 = 0$? This is an inconsistency, so again there is no unique solution.

For a square system, the only way to have a unique solution is to have linearly independent rows:

- If A is non-singular, then A^{-1} exists.
- If A^{-1} exists, then $x^* = A^{-1}d$ can be found.

4 Rank of a Matrix

The number of linearly independent rows of a matrix is also called its **rank**.

To find the rank of a matrix we may have to transform it into what is called an *echelon matrix* by using one of three types of **elementary row operations**:

1. Interchange any two rows.
2. Multiply (or divide) a row by any scalar $k \neq 0$.
3. Add “ k times any row” to another row.

Although these operations change the matrix into a different form, *none of them alter the rank*.

Example: Find the rank of matrix $A = \begin{bmatrix} 0 & -11 & -4 \\ 2 & 6 & 2 \\ 4 & 1 & 0 \end{bmatrix}$

Step 1: Check the first column for zeros. If any row has a zero in the first column, move that row to the bottom: $R_1 \circlearrowleft R_3$

$$A_1 = \begin{bmatrix} 4 & 1 & 0 \\ 2 & 6 & 2 \\ 0 & -11 & -4 \end{bmatrix}$$

Step 2: Transform the first column of A_1 into a unit vector $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. First, scale R_1 :

$$R_1 = \frac{1}{4}R_1 \quad \Rightarrow \quad A_2 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 2 & 6 & 2 \\ 0 & -11 & -4 \end{bmatrix}$$

Next, eliminate the 2 below the pivot (the 1 in R_1C_1):

$$R_2 = -2R_1 + R_2 \Rightarrow A_3 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & \frac{11}{2} & 2 \\ 0 & -11 & -4 \end{bmatrix}$$

Step 3: Scale the $\frac{11}{2}$ in A_3 to make it equal to 1:

$$R_2 = \frac{2}{11}R_2 \Rightarrow A_4 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & 1 & \frac{4}{11} \\ 0 & -11 & -4 \end{bmatrix}$$

Step 4: Eliminate the -11 in row 3:

$$R_3 = 11R_2 + R_3 \Rightarrow A_5 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & 1 & \frac{4}{11} \\ 0 & 0 & 0 \end{bmatrix}$$

We have now turned A into its *echelon* form (a triangular matrix), which satisfies:

1. Non-zero rows (i.e., with at least one non-zero element) appear above zero rows.
2. In every non-zero row, the first non-zero element is a 1.
3. The 1 in any row must be to the **left** of its counterpart in the immediately following row.

The rank of the matrix is equal to the number of non-zero rows, so A_5 has a rank of 2. Note that $r(A_5) = 2 = r(A)$, since the elementary row operations do not change the rank.

For a square $n \times n$ matrix, it must be that $r(A) = n$ for it to be non-singular. An $n \times n$ echelon matrix with no zero rows is non-singular; if $r(A) < n$, then A is singular and has no inverse.

Example: Find the rank of each matrix.

$$A = \begin{bmatrix} 1 & 5 & 1 \\ 0 & 3 & 9 \\ -1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & -1 & -4 \\ 3 & 1 & 2 \\ 6 & 1 & 0 \end{bmatrix}$$

For A :

$$\begin{aligned} R_3 &= R_1 + R_3 \Rightarrow \begin{bmatrix} 1 & 5 & 1 \\ 0 & 3 & 9 \\ 0 & 5 & 1 \end{bmatrix} & R_2 &= \frac{1}{3}R_2 \Rightarrow \begin{bmatrix} 1 & 5 & 1 \\ 0 & 1 & 3 \\ 0 & 5 & 1 \end{bmatrix} \\ R_3 &= 5R_2 + (-1)R_3 \Rightarrow \begin{bmatrix} 1 & 5 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix} & R_3 &= \frac{1}{14}R_3 \Rightarrow \begin{bmatrix} 1 & 5 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \\ & \Rightarrow r(A) = 3 \end{aligned}$$

For B :

$$\begin{aligned} R_1 &\circlearrowleft R_3 \Rightarrow \begin{bmatrix} 6 & 1 & 0 \\ 3 & 1 & 2 \\ 0 & -1 & -4 \end{bmatrix} & R_1 &= \frac{1}{6}R_1 \Rightarrow \begin{bmatrix} 1 & \frac{1}{6} & 0 \\ 3 & 1 & 2 \\ 0 & -1 & -4 \end{bmatrix} \\ R_2 &= 3R_1 + (-1)R_2 \Rightarrow \begin{bmatrix} 1 & \frac{1}{6} & 0 \\ 0 & \frac{-1}{2} & -2 \\ 0 & -1 & -4 \end{bmatrix} & R_2 &= -2R_2 \Rightarrow \begin{bmatrix} 1 & \frac{1}{6} & 0 \\ 0 & 1 & 4 \\ 0 & -1 & -4 \end{bmatrix} \\ R_3 &= R_2 + R_3 \Rightarrow \begin{bmatrix} 1 & \frac{1}{6} & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix} \\ & \Rightarrow r(B) = 2 \end{aligned}$$

Since $r(B) = 2 < 3$, B is not non-singular, so B has no inverse.

5 Test of Non-Singularity by Use of the Determinant

To determine non-singularity we can make use of the concept of the *determinant*.

Denote the determinant of A as $|A|$ (note: this is not the absolute value). For a 2×2 matrix $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, its determinant is

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

Example:

$$A = \begin{bmatrix} 10 & 4 \\ 8 & 5 \end{bmatrix} \Rightarrow |A| = 5(10) - 4(8) = 50 - 32 = 18$$

$$B = \begin{bmatrix} 3 & 5 \\ 0 & -1 \end{bmatrix} \Rightarrow |B| = -3 + 0 = -3$$

Note that a determinant can be reduced to a number (a scalar), and that this number can be negative.

Now consider

$$C = \begin{bmatrix} 3 & 8 \\ 3 & 8 \end{bmatrix} = \begin{bmatrix} c'_1 \\ c'_2 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 2 & 6 \\ 8 & 24 \end{bmatrix} = \begin{bmatrix} d'_1 \\ d'_2 \end{bmatrix}$$

Both C and D are linearly dependent: for C , $c'_1 = c'_2$, and for D , $4d'_1 = d'_2$. Interestingly, their determinants are

$$|C| = 24 - 24 = 0 \quad |D| = 2(24) - 8(6) = 48 - 48 = 0$$

A matrix with linearly *dependent* rows will have a determinant of zero.

5.1 Evaluating a Third-Order Determinant

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. The determinant of A can be expressed as follows:

$$\begin{aligned} |A| &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= (a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32}) - (a_{12}a_{21}a_{33} - a_{12}a_{23}a_{31}) + (a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}) \end{aligned}$$

which is a scalar.

Example: Find the determinant of A and B .

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\begin{aligned} |A| &= 2 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 1 \cdot \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ &= 2(45 - 48) - (36 - 42) + 3(32 - 35) = 2(-3) - (-6) + 3(-3) = -6 + 6 - 9 = -9 \end{aligned}$$

$$B = \begin{bmatrix} -7 & 0 & 3 \\ 9 & 1 & 4 \\ 0 & 6 & 5 \end{bmatrix}$$

$$|B| = -7 \cdot \begin{vmatrix} 1 & 4 \\ 6 & 5 \end{vmatrix} - 0 \cdot \begin{vmatrix} 9 & 4 \\ 0 & 5 \end{vmatrix} + 3 \cdot \begin{vmatrix} 9 & 1 \\ 0 & 6 \end{vmatrix} = -7(5 - 24) + 0 + 3(54) = 133 + 162 = 295$$

5.2 Minors and Cofactors

To find the determinant of a 3×3 matrix we eliminated a row and a column and were left with a sub-determinant of A . For a_{11} , we call it the *minor* of a_{11} , or $|M_{11}|$:

$$|M_{11}| = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, \quad |M_{12}| = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, \quad |M_{13}| = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

We can also define a *cofactor*:

$$c_{ij} = (-1)^{i+j} |M_{ij}|$$

Note that the sign which goes in front is determined by $i + j$: if $i + j$ is even, add a “+”; if $i + j$ is odd, add a “−”.

Example: Consider $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$

The minor of a_{12} is $|M_{12}| = \begin{vmatrix} 6 & 4 \\ 3 & 1 \end{vmatrix} = 6 - 12 = -6$, so the cofactor of a_{12} is $(-1)^{1+2} |M_{12}| = -(-6) = 6$.

What is the cofactor of a_{23} ?

$$c_{23} = (-1)^{2+3} \begin{vmatrix} 9 & 8 \\ 3 & 2 \end{vmatrix} = -[18 - 24] = 6$$

5.3 Higher-Order Determinants

We can express a third-order determinant as

$$\begin{aligned} |A| &= a_{11}|M_{11}| - a_{12}|M_{12}| + a_{13}|M_{13}| \\ &= a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} \\ &= \sum_{j=1}^3 a_{1j}c_{1j} \quad (\text{using the first row}) \end{aligned}$$

We can expand this to a fourth-order determinant,

$$|B| = \sum_{j=1}^4 b_{1j}c_{1j}$$

where the sub-determinants are now 3×3 .

You can do the determinant expansion using *any* row or column. Since we can use any row or column, we should choose the easiest one (i.e., the one with the most 0's or 1's).

5.4 Basic Properties of Determinants

1. The **transpose** of a matrix does not affect the value of the determinant, e.g.,

$$\begin{vmatrix} 4 & 3 \\ 5 & 6 \end{vmatrix} = \begin{vmatrix} 4 & 5 \\ 3 & 6 \end{vmatrix}$$

2. Interchanging any two rows or columns will alter the *sign*, but not the numerical value, e.g.,

$$\begin{vmatrix} 0 & 1 & 3 \\ 2 & 5 & 7 \\ 3 & 0 & 1 \end{vmatrix} = 0 - (2 - 21) + 3(-15) = 19 - 45 = -26$$

$$C_1 \circlearrowleft C_3 \Rightarrow \begin{vmatrix} 3 & 1 & 0 \\ 7 & 5 & 2 \\ 1 & 0 & 3 \end{vmatrix} = 3(15) - (21 - 2) + 0 = 45 - 19 = +26$$

3. Multiplication of any one row or column by a scalar k will change the value of the determinant k -fold. (Note this is *not* the same as multiplying the entire matrix by a scalar.)

$$\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = kad - kbc = k(ad - bc) = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

This also allows us to factor, e.g.,

$$\begin{vmatrix} 15a & 7b \\ 12c & 2d \end{vmatrix} \rightarrow \text{factor out a 3 from } C_1 \Rightarrow 3 \begin{vmatrix} 5a & 7b \\ 4c & 2d \end{vmatrix}$$

Then factor out a 2 from R_2 :

$$\Rightarrow 3(2) \begin{vmatrix} 5a & 7b \\ 2c & d \end{vmatrix} = 6(5ad - 14bc)$$

4. The addition (subtraction) of a multiple of any row to (from) another row will leave the value of the determinant unaltered. The same goes for column operations, e.g.,

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \rightarrow R_2 = R_2 + kR_1 \implies \begin{vmatrix} a & b \\ c + ka & d + kb \end{vmatrix}$$

$$\begin{aligned}
&= a(d + kb) - b(c + ka) = ad + akb - bc - bka \\
&= ad - bc = \begin{vmatrix} a & b \\ c & d \end{vmatrix}
\end{aligned}$$

5. If one row (or column) is a multiple of another row (or column), the **determinant** is 0, e.g.,

$$\begin{vmatrix} 2a & 2b \\ a & b \end{vmatrix} = 2ab - 2ab = 0$$

This is also called a “vanishing determinant.”

These properties can help simplify a determinant before evaluating it.

5.5 The Determinantal Criterion for Non-Singularity

Linear dependence implies a vanishing determinant. Conversely, linear independence implies $|A| \neq 0$, i.e., non-singularity. Putting it all together:

- $|A| \neq 0 \iff A$ has row (column) independence $\iff A$ is non-singular $\iff A^{-1}$ exists $\iff$ a unique solution $x^* = A^{-1}d$ exists.

Example: Does this system of equations have a unique solution?

$$\begin{aligned}
7x_1 - 3x_2 - 3x_3 &= 7 \\
2x_1 + 4x_2 + x_3 &= 0 \\
-2x_2 - x_3 &= 2
\end{aligned}$$

$$\begin{aligned}
A &= \begin{bmatrix} 7 & -3 & -3 \\ 2 & 4 & 1 \\ 0 & -2 & -1 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} 7 & -3 & -3 \\ 2 & 4 & 1 \\ 0 & -2 & -1 \end{vmatrix} \\
&= 7 \cdot \begin{vmatrix} 4 & 1 \\ -2 & -1 \end{vmatrix} - (-3) \cdot \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} + (-3) \cdot \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix} = 7(-4 + 2) + 3(-2) - 3(-4) \\
&= -14 - 6 + 12 = -8 \neq 0
\end{aligned}$$

Since $|A| \neq 0$, a unique solution exists.

6 Expansion of a Determinant by Alien Cofactors

Another property (a sixth property, if you like): the expansion of a determinant by *alien* cofactors (i.e., the cofactors of a “wrong” row) always yields a value of zero.

Example: Consider $A = \begin{bmatrix} 4 & 1 & 2 \\ 5 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$

Using the correct cofactor expansion:

$$\begin{aligned} |A| &= 4[2(3) - 0] - [5(3) - 1] + 2[0 - 2] = 4(6) - 14 - 4 = 6 \\ &= a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} \end{aligned}$$

What if we used *alien cofactors*? Suppose we used the cofactors of the **second**-row elements:

$$\begin{aligned} &a_{11}c_{21} + a_{12}c_{22} + a_{13}c_{23} \\ &= -4 \begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix} + \begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix} - 2 \begin{vmatrix} 4 & 1 \\ 1 & 0 \end{vmatrix} \\ &= -4(3) + (12 - 2) - 2(-1) = -12 + 10 + 2 = 0 \end{aligned}$$

More generally,

$$\begin{aligned} \sum_{j=1}^3 a_{1j}c_{2j} &= -a_{11} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{12} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{13} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ &= -a_{11}[a_{12}a_{33} - a_{13}a_{32}] + a_{12}[a_{11}a_{33} - a_{13}a_{31}] - a_{13}[a_{11}a_{32} - a_{12}a_{31}] \\ &= 0 \end{aligned}$$

since every term cancels. This applies to any alien cofactor expansion. Generally,

$$\sum_{j=1}^n a_{ij}c_{i'j} = 0 \quad \text{for } i \neq i'$$

Note the $i \neq i'$: in a regular expansion (where the i 's and the c_{ij} 's are consistent) we get $|A|$, whereas with alien cofactors everything cancels to zero.

7 Finding the Inverse of a Matrix

Assume

$$A_{n \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad \text{with } |A| \neq 0$$

Since each element of A has a cofactor, it is possible to form a matrix of cofactors,

$$C = [c_{ij}]$$

However, we will need the *transpose* of C , written C' and called the **adjoint** of A :

$$C'_{n \times n} = \text{Adj}(A) = \begin{bmatrix} c_{11} & c_{21} & \cdots & c_{n1} \\ c_{12} & c_{22} & \cdots & c_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ c_{1n} & c_{2n} & \cdots & c_{nn} \end{bmatrix}$$

Note that A and C' are conformable for multiplication. Their product AC' is another $n \times n$ matrix in which each element is a sum of products:

$$AC'_{n \times n} = \begin{bmatrix} \sum_{j=1}^n a_{1j}c_{1j} & \sum_{j=1}^n a_{1j}c_{2j} & \cdots & \sum_{j=1}^n a_{1j}c_{nj} \\ \sum_{j=1}^n a_{2j}c_{1j} & \sum_{j=1}^n a_{2j}c_{2j} & \cdots & \sum_{j=1}^n a_{2j}c_{nj} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=1}^n a_{nj}c_{1j} & \sum_{j=1}^n a_{nj}c_{2j} & \cdots & \sum_{j=1}^n a_{nj}c_{nj} \end{bmatrix}$$

Notice that all but the diagonal elements are *alien cofactor expansions*, which are all zeros. The diagonal elements use the cofactors of their own row, which give us exactly the determinant of A :

$$AC'_{n \times n} = \begin{bmatrix} |A| & 0 & \cdots & 0 \\ 0 & |A| & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & |A| \end{bmatrix}$$

Since $|A|$ is a non-zero scalar, we can factor it out:

$$AC' = |A| \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = |A| \cdot I_n$$

We are left with the determinant of A times an identity matrix. Then,

$$AC' = |A| \cdot I_n \Rightarrow \frac{AC'}{|A|} = I_n \Rightarrow A \cdot \frac{C'}{|A|} = I_n$$

Now pre-multiply both sides by A^{-1} :

$$A^{-1}A \cdot \frac{C'}{|A|} = A^{-1}I_n$$

$$\boxed{\frac{C'}{|A|} = A^{-1}}$$

That is, $\frac{C'}{|A|}$ (the adjoint divided by the determinant) is the inverse of A .

7.1 Steps to Find the Inverse

Step 1: Confirm $|A| \neq 0$.

Step 2: Find the cofactors of all elements of A and arrange them as a matrix $C = \begin{bmatrix} c_{ij} \end{bmatrix}$.

Step 3: Take the transpose of C to get $Adj(A) = C'$.

Step 4: Divide $Adj(A)$ by the determinant $|A|$.

Example: Find the inverse of $A = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

Step 1: $|A| = 3(0) - 2(1) = -2 \neq 0 \checkmark$

Step 2: Construct the matrix of cofactors (don't forget about the signs!):

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -2 & 3 \end{bmatrix}$$

Step 3: Take the transpose of C :

$$C' = \begin{bmatrix} 0 & -2 \\ -1 & 3 \end{bmatrix} = \text{Adj}(A)$$

Step 4: Divide by the determinant:

$$A^{-1} = \frac{\text{Adj}(A)}{|A|} = \frac{-1}{2} \begin{bmatrix} 0 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{-3}{2} \end{bmatrix}$$

Example: Find the inverse of $B = \begin{bmatrix} 4 & 1 & -1 \\ 0 & 3 & 2 \\ 3 & 0 & 7 \end{bmatrix}$

Step 1: Using the second row (it has the most zeros):

$$|B| = 0 + 3 \begin{vmatrix} 4 & -1 \\ 3 & 7 \end{vmatrix} - 2 \begin{vmatrix} 4 & 1 \\ 3 & 0 \end{vmatrix} = 3[28 + 3] - 2[0 - 3] = 93 + 6 = 99 \neq 0 \checkmark$$

Step 2: Cofactor matrix:

$$c_{11} = 21$$

$$c_{12} = 6$$

$$c_{13} = -9$$

$$c_{21} = -7$$

$$c_{22} = 31$$

$$c_{23} = 3$$

$$c_{31} = 5$$

$$c_{32} = -8$$

$$c_{33} = 12$$

$$C = \begin{bmatrix} 21 & 6 & -9 \\ -7 & 31 & 3 \\ 5 & -8 & 12 \end{bmatrix}$$

Step 3: Transpose to get the adjoint:

$$C' = \text{Adj}(B) = \begin{bmatrix} 21 & -7 & 5 \\ 6 & 31 & -8 \\ -9 & 3 & 12 \end{bmatrix}$$

Step 4: Divide by the determinant:

$$B^{-1} = \frac{Adj(B)}{|B|} = \frac{1}{99} \begin{bmatrix} 21 & -7 & 5 \\ 6 & 31 & -8 \\ -9 & 3 & 12 \end{bmatrix}$$

7.2 Solving Systems of Equations by Matrix Inversion

Example: Solve the system of equations

$$4x + 3y = 28$$

$$2x + 5y = 42$$

In matrix form:

$$\underbrace{\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 28 \\ 42 \end{bmatrix}}_d$$

Step 1: $|A| = 20 - 6 = 14 \neq 0 \checkmark$

Step 2: $C = \begin{bmatrix} 5 & -2 \\ -3 & 4 \end{bmatrix}$

Step 3: $C' = Adj(A) = \begin{bmatrix} 5 & -3 \\ -2 & 4 \end{bmatrix}$

Step 4: $A^{-1} = \begin{bmatrix} \frac{5}{14} & \frac{-3}{14} \\ \frac{-2}{14} & \frac{4}{14} \end{bmatrix}$

Then, since $A^{-1}Ax = A^{-1}d \Rightarrow x = A^{-1}d$:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{5}{14} & \frac{-3}{14} \\ \frac{-2}{14} & \frac{4}{14} \end{bmatrix} \begin{bmatrix} 28 \\ 42 \end{bmatrix} = \begin{bmatrix} \frac{5}{14} \cdot 28 - \frac{3}{14} \cdot 42 \\ -\frac{2}{14} \cdot 28 + \frac{4}{14} \cdot 42 \end{bmatrix} = \begin{bmatrix} 10 - 9 \\ -4 + 12 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 8$$

Example: Solve the system of equations

$$\begin{aligned} 4x_1 + x_2 - 5x_3 &= 8 \\ -2x_1 + 3x_2 + x_3 &= 12 \\ 3x_1 - x_2 + 4x_3 &= 5 \end{aligned}$$

In matrix form:

$$\underbrace{\begin{bmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 8 \\ 12 \\ 5 \end{bmatrix}}_d$$

Step 1: $|A| = 4[13] - [-11] + (-5)[-7] = 52 + 11 + 35 = 98 \neq 0 \checkmark$

Step 2: Cofactors:

$$\begin{array}{lll} c_{11} = 13 & c_{12} = 11 & c_{13} = -7 \\ c_{21} = 1 & c_{22} = 31 & c_{23} = 7 \\ c_{31} = 16 & c_{32} = 6 & c_{33} = 14 \end{array}$$

$$C = \begin{bmatrix} 13 & 11 & -7 \\ 1 & 31 & 7 \\ 16 & 6 & 14 \end{bmatrix}$$

Step 3: $C' = Adj(A) = \begin{bmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{bmatrix}$

Step 4: $A^{-1} = \frac{1}{|A|} \cdot Adj(A) = \begin{bmatrix} \frac{13}{98} & \frac{1}{98} & \frac{16}{98} \\ \frac{11}{98} & \frac{31}{98} & \frac{6}{98} \\ \frac{-7}{98} & \frac{7}{98} & \frac{14}{98} \end{bmatrix}$

Then,

$$x = A^{-1}d = \begin{bmatrix} \frac{13}{98} & \frac{1}{98} & \frac{16}{98} \\ \frac{11}{98} & \frac{31}{98} & \frac{6}{98} \\ \frac{-7}{98} & \frac{7}{98} & \frac{14}{98} \end{bmatrix} \begin{bmatrix} 8 \\ 12 \\ 5 \end{bmatrix} = \frac{1}{98} \begin{bmatrix} 104 + 12 + 80 \\ 88 + 372 + 30 \\ -56 + 84 + 70 \end{bmatrix} = \frac{1}{98} \begin{bmatrix} 196 \\ 490 \\ 98 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

8 Cramer's Rule

We have that $x^* = A^{-1}d = \frac{1}{|A|} \cdot \text{Adj}(A) \cdot d$, i.e.,

$$\begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} c_{11} & c_{21} & \cdots & c_{n1} \\ c_{12} & c_{22} & \cdots & c_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ c_{1n} & c_{2n} & \cdots & c_{nn} \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{bmatrix}$$

$n \times n \qquad \qquad \qquad n \times 1$

The resulting matrix will be $n \times 1$:

$$\begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} d_1 c_{11} + d_2 c_{21} + \cdots + d_n c_{n1} \\ d_1 c_{12} + d_2 c_{22} + \cdots + d_n c_{n2} \\ \vdots \\ d_1 c_{1n} + d_2 c_{2n} + \cdots + d_n c_{nn} \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} \sum_{i=1}^n d_i c_{i1} \\ \sum_{i=1}^n d_i c_{i2} \\ \vdots \\ \sum_{i=1}^n d_i c_{in} \end{bmatrix}$$

If we equate the corresponding elements,

$$x_1^* = \frac{1}{|A|} \sum_{i=1}^n d_i c_{i1}$$

$$x_2^* = \frac{1}{|A|} \sum_{i=1}^n d_i c_{i2}$$

$$\vdots$$

$$x_n^* = \frac{1}{|A|} \sum_{i=1}^n d_i c_{in}$$

These terms are similar to the Laplace expansion we would use to find $|A|$ using the first column,

$$\sum_{i=1}^n a_{i1} c_{i1},$$

only *replacing* the first column of A with the vector d , but keeping all other columns intact. Let's call this new determinant $|A_1| = \sum_{i=1}^n d_i c_{i1}$. Then,

$$x_1^* = \frac{|A_1|}{|A|}$$

Similarly, if we replace the *second* column of A by the column vector d (retaining all other columns), $|A_2| = \sum_{i=1}^n d_i c_{i2}$ and

$$x_2^* = \frac{|A_2|}{|A|}$$

We can generalize this procedure. To find the solution for the j^{th} variable x_j^* :

Step 1: Replace the j^{th} column of the determinant $|A|$ by the constant terms $d_1, d_2, \dots, d_n$.

Step 2: Get the new determinant $|A_j|$.

Step 3: Divide $|A_j|$ by $|A|$:

$$x_j^* = \frac{|A_j|}{|A|} = \frac{1}{|A|} \begin{vmatrix} a_{11} & a_{12} & \cdots & d_1 & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & d_2 & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & d_n & \cdots & a_{nn} \end{vmatrix}$$

This is known as **Cramer's Rule** (after Gabriel Cramer, 1704–1752). Note that matrix inversion gives us the solutions to all of the x^* 's at once, whereas with Cramer's rule you have to go one variable at a time.

Example: Use Cramer's rule to solve the following system of equations.

$$\begin{aligned} 5x_1 + 3x_2 &= 30 \\ 6x_1 - 2x_2 &= 8 \end{aligned} \quad \Rightarrow \quad \underbrace{\begin{bmatrix} 5 & 3 \\ 6 & -2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 30 \\ 8 \end{bmatrix}}_d$$

$$|A| = -10 - 18 = -28$$

$$|A_1| = \begin{vmatrix} 30 & 3 \\ 8 & -2 \end{vmatrix} = -60 - 24 = -84 \quad \Rightarrow \quad x_1^* = \frac{-84}{-28} = 3$$

$$|A_2| = \begin{vmatrix} 5 & 30 \\ 6 & 8 \end{vmatrix} = 40 - 180 = -140 \quad \Rightarrow \quad x_2^* = \frac{-140}{-28} = 5$$

Example: Use Cramer's rule to solve the following system of equations.

$$7x_1 - x_2 - x_3 = 0$$

$$10x_1 - 2x_2 + x_3 = 8$$

$$6x_1 + 3x_2 - 2x_3 = 7$$

$$|A| = \begin{vmatrix} 7 & -1 & -1 \\ 10 & -2 & 1 \\ 6 & 3 & -2 \end{vmatrix} = 7(4 - 3) + (-20 - 6) - (30 + 12) = 7 - 26 - 42 = -61$$

$$|A_1| = \begin{vmatrix} 0 & -1 & -1 \\ 8 & -2 & 1 \\ 7 & 3 & -2 \end{vmatrix} = 0 + (-16 - 7) - (24 + 14) = -23 - 38 = -61$$

$$|A_2| = \begin{vmatrix} 7 & 0 & -1 \\ 10 & 8 & 1 \\ 6 & 7 & -2 \end{vmatrix} = 7(-16 - 7) - 0 - (70 - 48) = -161 - 22 = -183$$

$$|A_3| = \begin{vmatrix} 7 & -1 & 0 \\ 10 & -2 & 8 \\ 6 & 3 & 7 \end{vmatrix} = 7(-14 - 24) + (70 - 48) + 0 = -266 + 22 = -244$$

$$x_1^* = \frac{-61}{-61} = 1 \quad x_2^* = \frac{-183}{-61} = 3 \quad x_3^* = \frac{-244}{-61} = 4$$

9 Finding Inverses by Row Operations

Recall our elementary row operations:

1. Interchange any two rows.
2. Multiply each element of a row by any scalar $k \neq 0$.
3. Add k times any row to another row.

9.1 Steps to Invert an $n \times n$ Matrix

Step 1: Form the $n \times 2n$ matrix $[A : I]$; that is, write down the n columns of A followed by the n columns of the identity matrix I .

Step 2: Apply elementary row operations until you end up with an $n \times 2n$ matrix $[I : B]$ whose **first n columns** are the columns of the identity matrix I .

Step 3: Then $B = A^{-1}$.

If it is impossible to perform such operations, then A has no inverse.

Example: Find the inverse of $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 3 \end{bmatrix}$

Will there be an inverse? $|A| = (9 - 16) - 3(3 - 4) + 3(4 - 3) = -7 + 3 + 3 = -1 \neq 0$
✓

$$[A : I] = \begin{bmatrix} 1 & 3 & 3 & : & 1 & 0 & 0 \\ 1 & 3 & 4 & : & 0 & 1 & 0 \\ 1 & 4 & 3 & : & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 = R_1 - R_3 \Rightarrow \begin{bmatrix} 1 & 3 & 3 & : & 1 & 0 & 0 \\ 1 & 3 & 4 & : & 0 & 1 & 0 \\ 0 & -1 & 0 & : & 1 & 0 & -1 \end{bmatrix}$$

$$R_2 = R_1 - R_2 \Rightarrow \begin{bmatrix} 1 & 3 & 3 & : & 1 & 0 & 0 \\ 0 & 0 & -1 & : & 1 & -1 & 0 \\ 0 & -1 & 0 & : & 1 & 0 & -1 \end{bmatrix}$$

$$R_1 = 3R_2 + R_1 \Rightarrow \begin{bmatrix} 1 & 3 & 0 & : & 4 & -3 & 0 \\ 0 & 0 & -1 & : & 1 & -1 & 0 \\ 0 & -1 & 0 & : & 1 & 0 & -1 \end{bmatrix}$$

$$R_2 \circlearrowleft R_3 \Rightarrow \begin{bmatrix} 1 & 3 & 0 & : & 4 & -3 & 0 \\ 0 & -1 & 0 & : & 1 & 0 & -1 \\ 0 & 0 & -1 & : & 1 & -1 & 0 \end{bmatrix}$$

$$\begin{aligned}
R_2 = -1 \cdot R_2 \\
R_3 = -1 \cdot R_3
\end{aligned}
\Rightarrow
\begin{bmatrix}
1 & 3 & 0 & : & 4 & -3 & 0 \\
0 & 1 & 0 & : & -1 & 0 & 1 \\
0 & 0 & 1 & : & -1 & 1 & 0
\end{bmatrix}$$

$$R_1 = -3R_2 + R_1 \Rightarrow
\begin{bmatrix}
1 & 0 & 0 & : & 7 & -3 & -3 \\
0 & 1 & 0 & : & -1 & 0 & 1 \\
0 & 0 & 1 & : & -1 & 1 & 0
\end{bmatrix}$$

$$\Rightarrow A^{-1} =
\begin{bmatrix}
7 & -3 & -3 \\
-1 & 0 & 1 \\
-1 & 1 & 0
\end{bmatrix}$$

If we had the column vector d , we could now find $x^* = A^{-1}d$.

Example: Find the inverse using row operations.

$$\begin{aligned}
\text{(i) } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} &\Rightarrow A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \\
\text{(ii) } B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} &\text{(Try this one on your own.)}
\end{aligned}$$

Example: Solve this system of equations using row operations to find the inverse.

$$\begin{aligned}
2x + 4y - 2z &= 6 \\
4x + 5y - 3z &= 13 \\
6x + 7y - 5z &= 19
\end{aligned}
\Rightarrow A = \begin{bmatrix} 2 & 4 & -2 \\ 4 & 5 & -3 \\ 6 & 7 & -5 \end{bmatrix}$$

$$R_1 = \frac{1}{2}R_1 \Rightarrow
\begin{bmatrix}
1 & 2 & -1 & : & \frac{1}{2} & 0 & 0 \\
4 & 5 & -3 & : & 0 & 1 & 0 \\
6 & 7 & -5 & : & 0 & 0 & 1
\end{bmatrix}$$

$$R_2 = R_2 - 4R_1 \Rightarrow
\begin{bmatrix}
1 & 2 & -1 & : & \frac{1}{2} & 0 & 0 \\
0 & -3 & 1 & : & -2 & 1 & 0 \\
6 & 7 & -5 & : & 0 & 0 & 1
\end{bmatrix}$$

$$R_3 = R_3 - 6R_1 \Rightarrow \begin{bmatrix} 1 & 2 & -1 & : & \frac{1}{2} & 0 & 0 \\ 0 & -3 & 1 & : & -2 & 1 & 0 \\ 0 & -5 & 1 & : & -3 & 0 & 1 \end{bmatrix}$$

$$R_2 = -\frac{1}{3}R_2 \Rightarrow \begin{bmatrix} 1 & 2 & -1 & : & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & : & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & -5 & 1 & : & -3 & 0 & 1 \end{bmatrix}$$

$$R_3 = R_3 + 5R_2 \Rightarrow \begin{bmatrix} 1 & 2 & -1 & : & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & : & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & -\frac{2}{3} & : & \frac{1}{3} & -\frac{5}{3} & 1 \end{bmatrix}$$

$$R_3 = -\frac{3}{2}R_3 \Rightarrow \begin{bmatrix} 1 & 2 & -1 & : & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & : & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & : & -\frac{1}{2} & \frac{5}{2} & -\frac{3}{2} \end{bmatrix}$$

$$\begin{aligned} R_2 &= R_2 + \frac{1}{3}R_3 \\ R_1 &= R_1 + R_3 \end{aligned} \Rightarrow \begin{bmatrix} 1 & 2 & 0 & : & 0 & \frac{5}{2} & -\frac{3}{2} \\ 0 & 1 & 0 & : & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & : & -\frac{1}{2} & \frac{5}{2} & -\frac{3}{2} \end{bmatrix}$$

$$R_1 = R_1 - 2R_2 \Rightarrow \begin{bmatrix} 1 & 0 & 0 & : & -1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & : & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & : & -\frac{1}{2} & \frac{5}{2} & -\frac{3}{2} \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{2} \begin{bmatrix} -2 & 3 & -1 \\ 1 & 1 & -1 \\ -1 & 5 & -3 \end{bmatrix}$$

Then,

$$x^* = A^{-1}d = \frac{1}{2} \begin{bmatrix} -2 & 3 & -1 \\ 1 & 1 & -1 \\ -1 & 5 & -3 \end{bmatrix} \begin{bmatrix} 6 \\ 13 \\ 19 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -12 + 39 - 19 \\ 6 + 13 - 19 \\ -6 + 65 - 57 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 8 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow x = 4, y = 0, z = 1$$

10 Homogeneous Equation Systems

Consider the system $Ax = 0$, where 0 is a vector. This is called a **homogeneous equation system**.

If A is non-singular, a homogeneous equation system can only yield the *trivial solution*

$$x_1^* = x_2^* = x_3^* = \cdots = x_n^* = 0.$$

This follows from

$$x^* = A^{-1}d \Rightarrow x^* = A^{-1} \underset{(n \times 1)}{0} = \underset{(n \times 1)}{0}$$

It can also be derived by Cramer's rule:

$$x_j^* = \frac{|A_j|}{|A|} = \frac{0}{|A|} = 0 \quad \forall j = 1, \dots, n$$

since replacing any column of A with a column of zeros produces a vanishing determinant.

Curiously, the only way to get a *non-trivial* solution is to have a *singular* coefficient matrix. But then

$$x_j^* = \frac{|A_j|}{|A|} = \frac{0}{0}$$

and the expression is undefined, so we can't use Cramer's rule. This means there isn't a unique solution but an infinite number of them.

To see this, consider

$$a_{11}x_1 + a_{12}x_2 = 0$$

$$a_{21}x_1 + a_{22}x_2 = 0$$

Clearly, $x_1^* = x_2^* = 0$ is a solution. However, if $|A| = 0$, then one equation is redundant, and

$$x_1^* = \left(-\frac{a_{12}}{a_{11}}\right)x_2^*$$

so there is an infinite number of solutions: for every x_2 there is an x_1 .